

Project Scope

Agentic AI Enabled School Transportation Safety & Compliance Platform

Abu Dhabi School Transportation Network

1. Project Overview

The objective of this project is to develop an intelligent school transportation management platform that combines IoT, Edge AI, Agentic AI, GPS tracking, safety monitoring, regulatory compliance automation, and operational optimization.

The platform will provide real-time visibility of school transportation operations while enabling autonomous decision-making and workflow execution through AI agents.

The solution will support:

- Real-time student transportation monitoring
- Automated safety incident detection
- Regulatory compliance enforcement
- Driver training and permit management
- Route optimization
- Operational workflow automation
- Integration with government permit systems and control centers

The platform will leverage Edge AI devices installed in buses to perform local inference and event detection while using centralized Agentic AI services to coordinate actions across the transportation ecosystem.

2. Business Objectives

Safety

- Improve student transportation safety.
- Reduce incident response time.
- Enable proactive risk detection.

Compliance

- Automate enforcement of transportation regulations.
- Ensure driver certification and training compliance.
- Integrate with government permit and licensing systems.

Operational Efficiency

- Reduce kilometers travelled.

- Improve bus utilization.
- Optimize route planning.
- Minimize manual intervention in transportation operations.

Cost Optimization

- Reduce cloud processing costs through edge inference.
 - Minimize communication bandwidth.
 - Automate operational workflows.
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3. Scope of Work

3.1 Real-Time Transportation Monitoring

Functional Scope

Capture and monitor:

- Bus location (GPS)
- Route progression
- Student pickup events
- Student drop-off events
- Bus occupancy
- Bus capacity utilization
- Driver information
- Trip status

Data Sources

- GPS devices
 - Student RFID/Smart Card systems
 - Mobile applications
 - Vehicle telematics
 - IoT sensors
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3.2 Edge AI Event Detection Platform

Edge Device Capabilities

Deploy AI-enabled edge devices within buses capable of:

- Local video analytics
- Sensor data processing
- Event classification
- Incident evidence generation

Data Sources

- ADAS cameras
- Accelerometers
- GPS
- Vehicle telemetry

Edge AI Use Cases

Collision Detection

Detect:

- Impact events
- Sudden deceleration
- Harsh braking
- Unsafe driving behavior

Driver Behavior Monitoring

Detect:

- Distracted driving
- Mobile phone usage
- Fatigue indicators
- Seatbelt violations

Passenger Safety Monitoring

Detect:

- Student left behind
- Unauthorized access
- Emergency situations

Output

Generate:

- Event type
- Confidence score
- Incident evidence
- Event timestamp
- GPS coordinates

Only relevant evidence and metadata will be transmitted to the central platform.

3.3 Agentic AI Safety Operations Layer

Objective

Replace manual monitoring and decision-making processes with autonomous AI agents.

Safety Monitoring Agent

Responsibilities:

- Monitor incoming incident streams.
- Classify incident severity.
- Determine response priority.
- Initiate response workflows.

Incident Response Agent

Responsibilities:

- Notify control center.
- Notify transportation coordinators.
- Notify school administrators.
- Escalate critical incidents.
- Track incident resolution.

Driver Compliance Agent

Responsibilities:

- Evaluate incident history.
- Assign mandatory training.
- Monitor training completion.
- Trigger compliance actions.

Regulatory Enforcement Agent

Responsibilities:

- Verify permit status.
 - Verify training status.
 - Apply enforcement rules.
 - Recommend driver suspension.
 - Recommend trip restrictions.
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3.4 Automated Compliance Management

Driver Training Workflow

Upon safety incident:

1. Incident detected.
2. Training requirement assigned.
3. Driver notified automatically.
4. Training completion monitored.
5. Compliance status tracked.

Compliance Actions

If training is not completed within 5 days:

- Driver suspension recommendation.
- Permit restriction workflow initiation.
- Control center notification.
- School transportation authority notification.

Integration Scope

Government permit systems:

- Driver permits
 - Operator licenses
 - Compliance records
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3.5 Incident Management Automation

Event Lifecycle

Detection → Validation → Notification → Investigation → Resolution → Reporting

Automated Actions

- Create incident record
- Generate evidence package
- Notify stakeholders
- Trigger investigation workflow
- Monitor SLA compliance
- Generate closure report

Notification Channels

- SMS
- Email
- Mobile App
- Control Center Dashboard

3.6 Route Optimization Platform

Objectives

Reduce:

- Total kilometers travelled
- Fuel consumption
- Travel time
- Fleet operating costs

Inputs

- Student locations
- Bus capacities
- Historical routes
- Traffic data
- School schedules
- Operational constraints

AI Capabilities

- Route optimization
- Dynamic rerouting
- Capacity balancing
- Fleet utilization optimization

Expected Outcomes

- Reduced transportation costs
 - Increased fleet efficiency
 - Improved student service levels
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3.7 Control Center Platform

Features

Real-time operational dashboard including:

- Fleet monitoring
- Incident monitoring
- Driver compliance status
- Route performance
- Safety analytics
- Alert management

Command Center Functions

- Incident review

- Escalation management
 - Manual override
 - Workflow monitoring
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3.8 Government System Integration

Integration Areas

Transportation authority systems:

- Driver permits
- Vehicle permits
- Compliance databases
- Fine management systems

Automated Transactions

- Permit verification
 - Driver status updates
 - Fine issuance requests
 - Compliance reporting
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4. Agentic AI Scope

AI Agents Included

Fleet Monitoring Agent

Monitors:

- Fleet health
- Route adherence
- Operational anomalies

Safety Agent

Monitors:

- Safety incidents
- Driver behavior
- Passenger safety events

Compliance Agent

Monitors:

- Training requirements
- Permit validity

- Regulatory obligations

Incident Response Agent

Executes:

- Notifications
- Escalations
- Resolution workflows

Route Optimization Agent

Executes:

- Route planning
- Dynamic route adjustments
- Capacity optimization

Executive Insights Agent

Provides:

- Operational summaries
 - Compliance KPIs
 - Risk indicators
 - Performance recommendations
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5. Non-Functional Requirements

Performance

- Real-time event detection
- Incident notification within seconds
- High availability operations

Scalability

Support:

- Thousands of buses
- Millions of trip records
- Concurrent edge devices

Security

- End-to-end encryption
- Secure device authentication
- Role-based access control

Reliability

- Offline edge operation
 - Event buffering
 - Guaranteed event delivery
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6. Expected Benefits

Safety Benefits

- Faster incident response
- Reduced safety risks
- Improved student protection

Compliance Benefits

- Automated regulatory enforcement
- Reduced manual administration
- Improved audit readiness

Operational Benefits

- Reduced route costs
- Improved fleet utilization
- Faster decision making

Financial Benefits

- Lower operational costs
 - Reduced cloud processing costs
 - Reduced bandwidth consumption
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7. Deliverables

1. Edge AI Bus Monitoring Solution
2. Central Transportation Operations Platform
3. Agentic AI Decision and Workflow Engine
4. Incident Management System
5. Compliance Automation System
6. Government Integration Framework
7. Route Optimization Engine
8. Control Center Dashboards
9. Mobile Notification Services
10. Analytics and Reporting Platform